Pulkit Chawla

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Summary

Software engineer (6+ years at Oracle Cloud and Qubole) turned AI engineer after an MSc in Data Science and AI at TU Eindhoven. In 2026 built production-grade agent systems: a contract-bound microbot harness that only ships verified AI-written code, a native macOS computer-use agent, and a personal email agent with structural prompt-injection defences. Owns services end to end: design, build, test, release, on-call.

Work authorisation: NL orientation-year permit (full labour-market access); HSM-eligible under the reduced graduate criterion

Experience

Freelance and personal projects — Jan. 2026 – Present
Independent AI / software engineer — *Eindhoven, Netherlands*

- Designed and shipped four agent systems (below): a contract-bound microbot harness for AI-written code, a native macOS computer-use agent, a personal email agent with structural prompt-injection defences, and a multi-account trading fan-out platform
- Built a WhatsApp AI assistant for an Eindhoven student food-catering business (Twilio, Gemini/Gemma, FastAPI on DigitalOcean) handling orders, meal selection, leave days and billing questions

Eindhoven University of Technology, IMAGEN Programme — Jan. 2025 – Nov. 2025
Graduate AI Researcher (MSc thesis), MLOps and Computer Vision — *Eindhoven, Netherlands*

- Built the MLOps platform behind IMAGEN, a Dutch government-funded animal-welfare research programme run by TU/e, Wageningen University and Hendrix Genetics, on SURF's Spider HPC cluster and dCache storage
- Redesigned the pipeline from a PiCaS pilot-job setup into a decoupled control plane: ClearML experiment tracking and a CouchDB job queue feeding SLURM jobs that run in Singularity containers, with a web UI to configure clusters, experiments and runs
- Wrote the FAIR data pipeline that registers uploaded datasets, converts them to HDF5 and catalogs them with provenance; at eight loader workers HDF5 delivered 217.4 images/s against 112.3 for raw images, near-linear scaling that cut GPU idle time during training
- Exposed the pipeline through a web UI and REST API so researchers could fine-tune detection models without logging into the HPC cluster; per-animal bounding boxes fed behavioural analysis such as feeding-order dominance

Oracle Cloud Infrastructure — Jan. 2021 – Aug. 2023
Member of Technical Staff, Big Data Service (BDS) — *Bangalore, India*

- Sole owner of the customer-facing console of OCI Big Data Service in a ~30-engineer team: built, tested, released and operated the Node.js UI used to manage Hadoop clusters, and trained other engineers to take it over before leaving
- Owned the Hive and HBase distributions: forked upstream open source, set up TeamCity pipelines that built and deployed every PR to staging automatically, and kept both patched by back-porting upstream CVE fixes, a hard requirement for enterprise customers leaving Cloudera
- Served in the production on-call rotation, resolving P0/P1 incidents with the US-based support and architecture teams

Qubole (acquired by Idera) — Jan. 2018 – Dec. 2020
Software Engineer → Software Engineer II, Cluster Orchestration — *Bangalore, India*

- One of three engineers owning the cluster-orchestration backend of Qubole's big-data platform, spinning up and managing Hadoop, Spark, Presto and Hive clusters inside customer accounts on AWS, Azure and OCI
- Added Datadog monitoring for cluster metrics and Jenkins QA pipelines gating every release on client-specific integration tests across staging and production
- Built secure tunnels between Qubole's control plane and clusters running in customers' own cloud accounts, letting customers operate their clusters directly from the UI

Projects & Research

devteam: contract-bound microbot harness for agent-written software | *Node.js* — Jul 2026 – present
github.com/thispc/devteam

- Every module is a contract (interface, tests, pinned toolchain) hashed into its identity; agent-written code is admitted only after passing in a network-less Docker container, run twice in two environments with transcripts diffed, plus held-out tests the author never sees
- Ran an end-to-end trial in which an AI manager hired its own team, drafted contracts from three examples and shipped gate-verified modules with zero human interventions; 356 commits, held-out vaults caught real bugs invisible to the visible suite
- Quarantined unverifiable effects (network, clocks, human input) into declared holes with owner-granted host allowlists judged on recorded real traffic; a headless real-mouse probe harness caught 15 bugs that unit tests had passed

HumanHands: native macOS computer-use agent | *Swift, SwiftUI* 2026
github.com/thispc/humanhands

- Agent that perceives the screen through the macOS Accessibility tree (local, pixel-exact, no screenshots to the model), reasons with Claude, and drives the real cursor along human-like paths with human typing cadence; vision kept only as a concurrent fallback
- Security spine: credentials live in the Keychain and reach the model only as placeholders substituted at the last instant; a domain-verified gate refuses to type a secret on an unconfirmed site; user-owned app and site allowlists bound its reach
- Fully native Swift across 10 SwiftPM targets with every OS primitive behind a protocol boundary; 135 tests run off-device

Personal email agent (NanoClaw fork) | *TypeScript, Node.js, Claude Agent SDK* 2026
github.com/thispc/nanoclaw

- Structural defence against prompt injection: a tool-less reader model is the only component that touches raw email, the acting agent sees only sanitised digests, and every outbound action stops at a one-tap approval that sends exactly the pre-created draft, enforced in host code rather than prompts
- Extended the open-source NanoClaw runtime with an iMessage channel and a portable plain-file knowledge base (+16k lines); runs 24/7 on own hardware across three daemons (iMessage, ingestion, PWA) at zero idle cost

jobhunt: agent-operated job-application system | *Python, FastAPI, Playwright* Sep 2026

- Directed and shipped, with Claude Code, a pipeline that polls nine applicant-tracking systems across 160+ Dutch companies, matches employers against the 12,900-row IND sponsor register, scores postings, tailors a LaTeX CV per job from an evidence-tagged profile, and fills application forms with Playwright behind a submit guard

LLM question answering with RAG (TU/e summer research) | *Python, LangChain* Jun – Aug 2024

- Built a question-answering pipeline over a corpus of 10,000+ academic documents using LangChain, the GPT-4 API and FAISS vector search with OpenAI embeddings, keeping retrieval under 100 ms per query
- Benchmarked against BM25 and TF-IDF baselines and tuned chunking, context management and prompting (chain-of-thought, few-shot) to beat plain LLM answers on domain-specific queries

Education

Eindhoven University of Technology (TU/e) Aug. 2023 – Nov. 2025
Master of Science in Data Science and Artificial Intelligence *Eindhoven, Netherlands*

- Thesis: A Reimagined and Implemented MLOps Architecture for Supercomputing Environments (IMAGEN programme, SURF Spider HPC), supervisors Dr. Yanja Dajsuren and Manu Agarwal

Birla Institute of Technology, Mesra Aug. 2014 – Jun. 2018
Bachelor of Engineering in Information Technology, CGPA 8.37/10, First Class with Distinction *Ranchi, India*

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Technical Skills

AI/ML: LLM agents (Claude Agent SDK), RAG and vector search (FAISS), prompt-injection defences, LangChain, PyTorch, computer vision (object detection, tracking), fine-tuning, model serving and inference services, ClearML
Languages: Python, TypeScript/JavaScript, Swift, Java, SQL, Bash
Backend & Tooling: FastAPI, Spring, Node.js, REST APIs, Jenkins, GitHub Actions, CI/CD, Datadog, Elasticsearch, Playwright
Big Data & Cloud: Hadoop, Spark, AWS, Azure, OCI, Docker, Kubernetes
Spoken languages: English: fluent; Hindi: native; Dutch: beginner, learning